

Sociotechnical Security

CS 8803/4803

Professor Veronica Rivera

Lecture 12 – Scams I (Problems)



Content warning: This lecture and discussion may surface themes related to emotional harm in romantic/close relationships

Wrapping up: usable security and privacy

- Core problem: Users make mistakes, which lead to insecure behavior
 - We can study these mental models through interviews, surveys, and lab-studies
- General approach: Problematize-model-fix
 - Authentication
 - Phishing
 - Warnings
 - Privacy
- Old papers have new applications:
 - E.g., early warnings research informing misinformation and AI content indicators

Digital safety (or trust and safety)

“Trust and Safety is the study of how the technology platforms that facilitate interactions between human beings can be used to cause harm and the steps that can be taken to prevent or mitigate those harms.”

- Alex Stamos, Shelby Grossman, and Riana Pfefferkorn

(Trust and Safety: An Open Source Textbook)

Questions people ask in digital safety

- How does harm happen?
- How do victims make sense of this?
- What strategies do they employ to try to mitigate these harms?
- What post-harm recourses do people have?
- Why do they use them (or not use them)?
- What tensions emerge between people's use of tech & harm/protective strategies?

Digital safety topics we'll discuss

- Scams
- Online hate and harassment
- Intimate partner violence
- At-risk user groups
- Digital intimacy
- Youth Safety
- Majority world considerations
- Ethics

Today:

- Types of scams studied in digital safety
- Approaches to studying scams
- How researchers think AI is being used in scam ecosystem
- Discussion on scams led by Heeyun

A note on grading

- Project proposal
 - Graded based on clarity and completion **not** feasibility and scope of project
 - Midpoint report will also be graded primarily on clarity and completion in addition to
 - **How well you have positioned your problem within prior academic work**
 - **Whether you have made changes to your project's scope and/or methods based on concerns I raised about this during the check-in meeting**
 - **Reminder:** Final project deliverables **will** be graded on feasibility, scope, and whether your results align with your methods/research questions
- Participation
 - Each class where we have a discussion is worth 10 points
 - 5 for showing up
 - 5 for engaging meaningfully in the discussion: contributing at least one comment/new discussion question/follow up question
 - Not enough to say “I agree/disagree with [insert students' name here]” → why do you agree or disagree?

Reminder: Content warning

We will cover real-world harmful behavior.

These topics include, but are not limited to: interpersonal abuse, sexual harassment/assault, hate speech, harassment, stalking, child and adult sexual exploitation, and more.

Reminder: Content warning

- I will post content warnings on papers that might be triggering for one of these reasons
- However, there are many reasons why a paper might be triggering to someone that I may not catch
- All students are encouraged to preview papers for upcoming classes

If you think one or both papers for an upcoming class would be distressing:

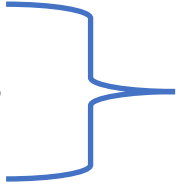
- You can use one of your drop days
- If you are out of drop days, email me **at least 72 hours in advance of the deadline** for an alternative reading
 - This applies to the leading discussion assignment as well
- If you feel uncomfortable during class discussion and need a break, you are welcomed to step out

Scams

A scam is a deceptive scheme designed to trick people into giving away their money or personal data

- Previously looked at: phishing

Many other types of scams studied in digital safety:

- Romance scams
 - Cryptocurrency scams
 - Advanced fee fraud
 - Scams targeting older adults
 - Tech support scams and robocalls
 - Sextortion
- 
- Pig-butchering scams

Many scams operate as organized industries

- Many scams depend on labor from trafficked or coerced workers in scam compounds
- Cyber-slavery:
 - Individuals are deceived with promises of lucrative jobs (scam)
 - Trafficked across borders
 - Forced to work in scam compounds running large-scale online fraud operations
 - Often in Southeast Asia (e.g., Myanmar)
- Recruiters exploit cultural ties to draw victims into scam work

Landscape of scam research in digital safety

How scams operate	Victim experiences and harms	Why people engage with scams	Protection and victim support
		Chanenson et al. (CHI 2026)	
	Bouma-Sims et al. (SOUPS 2024)		
	Oak & Shafiq (SOUPS 2025): Victims, Vigilantes, and Advice-Givers		
Miramirkhani et al. (NDSS 2017)			Deng et al. (CHI 2026)
Oak & Shafiq Hello, is this Anna? (SOUPS 2025)			